Chapter 8

Monads

The notion of monads originates in homological algebra and category theory.

8.1 Definition

Here are some notations for defining monads.[1]

- Any endofunctor $T : \mathcal{C} \longrightarrow \mathcal{C}$ has composites $T^2 = T \circ T : \mathcal{C} \longrightarrow \mathcal{C}$ and $T^3 = T^2 \circ T : \mathcal{C} \longrightarrow \mathcal{C}$.

- If $\mu : T^2 \xrightarrow{\cdot} T$ is a natural transformation, with components $\mu_x : T^2x \longrightarrow Tx$ for each $x \in \mathcal{X}$, then

  - $T\mu : T^3 \longrightarrow T^2$ denotes the natural transformation with components $(T\mu)_x = T(\mu x) : T^3x \longrightarrow T^2x$.

  - $\mu T : T^3 \longrightarrow T^2$ denotes the natural transformation with components $(\mu T)_x = \mu_{Tx} : T^3x \longrightarrow T^2x$.

[1] MacLane (1997, p.137)

Definition 8.1.1 (Monad[2])**.** A *monad* $T = \langle \boldsymbol{T}, \eta, \mu \rangle$ in $\mathcal{C}$ consists of a functor $\boldsymbol{T} : \mathcal{C} \longrightarrow \mathcal{C}$ and two natural transformations

$$\eta : \boldsymbol{id}_{\mathcal{C}} \stackrel{\cdot}{\longrightarrow} \boldsymbol{T}, \qquad \mu : \boldsymbol{T}^2 \stackrel{\cdot}{\longrightarrow} \boldsymbol{T}$$

which make the following diagrams commute for any object A in $\mathcal{C}$:

$$\begin{array}{ccc} \boldsymbol{T}^3 A & \xrightarrow{\boldsymbol{T}\mu_A} & \boldsymbol{T}^2 A \\ {\scriptstyle \mu_{\boldsymbol{T}A}}\downarrow & & \downarrow{\scriptstyle \mu_A} \\ \boldsymbol{T}^2 A & \xrightarrow[\mu_A]{} & \boldsymbol{T}A \end{array} \qquad \begin{array}{ccccc} \boldsymbol{T}A & \xrightarrow{\eta_{\boldsymbol{T}A}} & \boldsymbol{T}^2 A & \xleftarrow{\boldsymbol{T}\eta_A} & \boldsymbol{T}A \\ & \searrow^{\boldsymbol{id}_{\mathcal{C}}} & \downarrow{\scriptstyle \mu_A} & \swarrow_{\boldsymbol{id}_{\mathcal{C}}} & \\ & & \boldsymbol{T}A & & \end{array}$$

We call η the *unit* and μ the *multiplication* or the *join*.

The left diagram is *associativity law*, and the right diagram is *unitary law* where η plays a role of unit for multiplication μ.

Theorem 8.1.2. For any adjunction $\langle \boldsymbol{F}, \boldsymbol{G}, \eta, \varepsilon \rangle : \mathcal{X} \longrightarrow \mathcal{A}$,

$$\langle \boldsymbol{GF}, \eta : \boldsymbol{id}_{\mathcal{X}} \longrightarrow \boldsymbol{T}, G\varepsilon_{F_} : \boldsymbol{T}^2 \longrightarrow \boldsymbol{T} \rangle$$

is a monad. This is called a *monad defined by the adjunction.*

Proof. We show that the triple defined by a given adjunction as above qualifies as monad. Regarding the associativity law:

$$\begin{array}{rcll} \mu_A \circ \boldsymbol{T}\mu_A & = & \boldsymbol{G}\epsilon_{\boldsymbol{F}A} \circ \boldsymbol{GFG}\epsilon_{\boldsymbol{F}A} & \\ & = & \boldsymbol{G}(\epsilon_{\boldsymbol{F}A} \circ \boldsymbol{FG}\epsilon_{\boldsymbol{F}A}) & \\ & = & \boldsymbol{G}(\epsilon_{\boldsymbol{F}A} \circ \epsilon_{\boldsymbol{FGF}A}) & \text{(by naturality of } \epsilon) \\ & = & \boldsymbol{G}\epsilon_{\boldsymbol{F}A} \circ \boldsymbol{G}\epsilon_{\boldsymbol{FGF}A} & \\ & = & \mu_A \circ \mu_{\boldsymbol{T}A} & \end{array}$$

The unitary laws are proved as below by using the triangular identity of adjunctions:

$$\begin{array}{lllllllll} \mu_A \circ \eta_{\boldsymbol{T}A} & = & \boldsymbol{G}\epsilon_{\boldsymbol{F}A} \circ \eta_{\boldsymbol{GF}A} & = & id_{\boldsymbol{GF}A} & & & & \\ \mu_A \circ \boldsymbol{T}\eta_A & = & \boldsymbol{G}\epsilon_{\boldsymbol{F}A} \circ \boldsymbol{GF}\eta_A & = & \boldsymbol{G}(\epsilon_{\boldsymbol{F}A} \circ \boldsymbol{F}\eta_A) & = & \boldsymbol{G}(\boldsymbol{id}_{\boldsymbol{F}A}) & = & \boldsymbol{id}_{\boldsymbol{GF}A} \end{array}$$

□

[2]MacLane (1997, p.137:(1)(2))

Example 8.1.3. Recall the adjunction $_ \times A \dashv _^A$ where $\eta = \lambda(id_{_\times A})$ and $\epsilon = ev_{A,_}$. This adjunction defines a *state monad*.

8.2 $\boldsymbol{T}$-algebra

$\boldsymbol{T}$-algebra is the construction due to **?**.

Definition 8.2.1 ($\boldsymbol{T}$-algebra)**.** For any monad $\langle \boldsymbol{T}, \eta, \mu \rangle$ in a category $\mathcal{X}$, $\boldsymbol{T}$-algebra $\langle X, h \rangle$ is a pair consisting of an object $X \in ob\mathcal{X}$ and a morphism $h : \boldsymbol{T}X \longrightarrow X$ that makes the following diagrams commute:

$$\begin{array}{ccc} \boldsymbol{T}^2X & \xrightarrow{\boldsymbol{T}h} & \boldsymbol{T}X \\ {\scriptstyle \mu_X}\downarrow & & \downarrow{\scriptstyle h} \\ \boldsymbol{T}X & \xrightarrow[h]{} & X \end{array} \qquad \begin{array}{ccc} X & \xrightarrow{\eta_X} & \boldsymbol{T}X \\ & {\scriptstyle id_X}\searrow & \downarrow{\scriptstyle h} \\ & & X \end{array}$$

X and h are called *the underlying object* and *the structure map* of the algebra.

The left and right diagrams respectively represents the associative law and the unit law of the algebra.

Definition 8.2.2 (Cateogory of $\boldsymbol{T}$-algebra)**.** For any monad $\langle \boldsymbol{T}, \eta, \mu \rangle$ in a category $\mathcal{X}$, a category of $\boldsymbol{T}$-algebra (notation $X^{\boldsymbol{T}}$) consists of (as objects) all $\boldsymbol{T}$-algebras, and(as morphisms from $\langle X, h \rangle$ to $\langle X', h' \rangle$) $\mathcal{X}$-morphisms of the form $f : X \longrightarrow X'$ that makes the following diagram commute:

$$\begin{array}{ccc} X & \xleftarrow{h} & \boldsymbol{T}X \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle \boldsymbol{T}f} \\ X' & \xleftarrow{h'} & \boldsymbol{T}X' \end{array}$$

Identity morphisms and compositions are defined in a straightforward manner.

The folloiwng theorem shows a decomposition of a given monad into adjunction.

Theorem 8.2.3. For any monad $\langle \boldsymbol{T}, \eta, \mu \rangle$ in $\mathcal{X}$, there is an adjunction $\langle \boldsymbol{F}^{\boldsymbol{T}}, \boldsymbol{G}^{\boldsymbol{T}}, \eta^{\boldsymbol{T}}, \varepsilon^{\boldsymbol{T}} \rangle : \mathcal{X} \longrightarrow \mathcal{X}^{\boldsymbol{T}}$ where

- $F^{\boldsymbol{T}} : \mathcal{X} \longrightarrow \mathcal{X}^{\boldsymbol{T}}$ sends X to $\langle \boldsymbol{T}X, \mu_X \rangle$ and f to $\boldsymbol{T}$f.
- $G^{\boldsymbol{T}} : \mathcal{X}^{\boldsymbol{T}} \longrightarrow \mathcal{X}$ sends $\langle X, h \rangle$ to X and f to f.
- $\eta^{\boldsymbol{T}}$ is η.
- $\epsilon^{\boldsymbol{T}}$ sends $\langle X, h \rangle$ to h.

Proof. □

Theorem 8.2.4. The adjunction $\langle \boldsymbol{F}^{\boldsymbol{T}}, \boldsymbol{G}^{\boldsymbol{T}}, \eta^{\boldsymbol{T}}, \varepsilon^{\boldsymbol{T}} \rangle$ defines the original monad $\langle \boldsymbol{T}, \eta, \mu \rangle$.

Proof. □

8.3 Kleisli Category

Kleisli cateogry is the construction due to Kleisli (1965).

Definition 8.3.1 (Kleisli category)**.** For any monad $\langle \boldsymbol{T}, \eta, \mu \rangle$ in $\mathcal{X}$, the *Kleisli category* $\mathcal{X}_{\boldsymbol{T}}$ consists of (as objects) $X_{\boldsymbol{T}}$ in $\mathcal{X}_{\boldsymbol{T}}$ for each object X in $\mathcal{X}$, and (as morphism) $f^\flat : X_{\boldsymbol{T}} \longrightarrow Y_{\boldsymbol{T}}$ in $\mathcal{X}_{\boldsymbol{T}}$ for each morphism $f : X \longrightarrow \boldsymbol{T}Y$ in $\mathcal{X}$.
The composite of $f^\flat : X_{\boldsymbol{T}} \longrightarrow Y_{\boldsymbol{T}}$ with $g^\flat : Y_{\boldsymbol{T}} \longrightarrow Z_{\boldsymbol{T}}$ is defined by $g^\flat \circ f^\flat = (\eta_Z \circ \boldsymbol{T}g \circ f)^\flat$.

Theorem 8.3.2. For any monad $\langle \boldsymbol{T}, \eta, \mu \rangle$ and a Kleisli category $\mathcal{X}_{\boldsymbol{T}}$, functors $\boldsymbol{F_T} : \mathcal{X} \longrightarrow \mathcal{X}_{\boldsymbol{T}}$ and $\boldsymbol{G_T} : \mathcal{X}_T \longrightarrow \mathcal{X}$ are defined by

$$\begin{array}{lllll} \boldsymbol{F_T}: & k : X \longrightarrow Y & \mapsto & (\eta_y \circ k)^\flat : X_{\boldsymbol{T}} \longrightarrow Y_{\boldsymbol{T}} \\ \boldsymbol{G_T}: & f^\flat : X_{\boldsymbol{T}} \longrightarrow Y_{\boldsymbol{T}} & \mapsto & \eta_Y \circ \boldsymbol{T}f : \boldsymbol{T}X \longrightarrow \boldsymbol{T}^2 Y \longrightarrow \boldsymbol{T}Y \end{array}$$

respectively, so that $G_{\boldsymbol{T}} X_{\boldsymbol{T}} = \boldsymbol{T}X$ on objects.

There is an adjunction $\langle \boldsymbol{F_T}, \boldsymbol{G_T}, \eta_{\boldsymbol{T}}, \varepsilon_{\boldsymbol{T}} \rangle$ where $\eta_{\boldsymbol{T}} = \eta$ and $\varepsilon_{\boldsymbol{T}} = \eta$.

Proof. □

Theorem 8.3.3. The adjunction $\langle \boldsymbol{F_T}, \boldsymbol{G_T}, \eta_{\boldsymbol{T}}, \varepsilon_{\boldsymbol{T}} \rangle$ defines the original monad $\langle \boldsymbol{T}, \eta, \mu \rangle$.

Proof. □

8.4 Comparison Theorem

Theorem 8.4.1 (Comparison Theorem for $\boldsymbol{T}$-algebra)**.** Given an adjunction $\langle \boldsymbol{F}, \boldsymbol{G}, \eta, \varepsilon \rangle : \mathcal{X} \longrightarrow \mathcal{A}$ and $\langle \boldsymbol{GF}, \eta, G\varepsilon_{F_} \rangle$ is a monad that it defines in $\mathcal{X}$, there is a unique functor $\boldsymbol{K} : \mathcal{A} \longrightarrow \mathcal{X}^{\boldsymbol{T}}$ with $\boldsymbol{G^T K} = \boldsymbol{G}$ and $\boldsymbol{KF} = \boldsymbol{F^T}$.

Proof. □

Theorem 8.4.2 (Comparison Theorem for Kleisli Category)**.** Given an adjunction $\langle \boldsymbol{F}, \boldsymbol{G}, \eta, \varepsilon \rangle : \mathcal{X} \longrightarrow \mathcal{A}$ and $\langle \boldsymbol{GF}, \eta, G\varepsilon_{F_} \rangle$ is a monad that it defines in $\mathcal{X}$, there is a unique functor $\boldsymbol{L} : \mathcal{X}^{\boldsymbol{T}} \longrightarrow \mathcal{A}$ with $\boldsymbol{GL} = \boldsymbol{G_T}$ and $\boldsymbol{LF_T} = \boldsymbol{F}$.

Proof. □

Definition 8.4.3. Given a monad $\langle T, \eta, \mu \rangle$, there is a category with objects all adjunctions $\langle \boldsymbol{F}, \boldsymbol{G}, \eta, \varepsilon \rangle : \mathcal{X} \longrightarrow \mathcal{A}$ which defines $\langle T, \eta, \mu \rangle$ in $\mathcal{X}$, and with morphisms maps of adjunctions.

Theorem 8.4.4. The above category has an initial object $\langle \boldsymbol{F_T}, \boldsymbol{G_T}, \eta_{\boldsymbol{T}}, \varepsilon_{\boldsymbol{T}} \rangle$ and a terminal object $\langle \boldsymbol{F^T}, \boldsymbol{G^T}, \eta^{\boldsymbol{T}}, \varepsilon^{\boldsymbol{T}} \rangle$ with the comparison functor:

$$\mathcal{X}_{\boldsymbol{T}} \xrightarrow{\boldsymbol{L}} \mathcal{A} \xrightarrow{\boldsymbol{K}} \mathcal{X}^{\boldsymbol{T}}$$

8.5 Kleisli Triple

There is an alternative description of a monad (cf. Moggi (1989)). The *Kleisli triple*s is introduced by E. Manes in his 1976 book called "Algebraic Theories".

Definition 8.5.1 (Kleisli triples)**.** [3]A Kleisli triple over $\mathcal{X}$ is a system(?) consisting of functions:

(i) $\boldsymbol{T}$: $ob\mathcal{X} \longrightarrow ob\mathcal{X}$

(ii) η: $ob\mathcal{X} \longrightarrow mor\mathcal{X}$ with value $\eta_A : A \longrightarrow \boldsymbol{T}A$ on object A

(iii) $-^*$: a function on morphisms $f : A \longrightarrow \boldsymbol{T}B$ with value $f^* : \boldsymbol{T}A \longrightarrow \boldsymbol{T}B$

which is subject to the following equations:

(K1) $(\eta_A)^* = id_{\boldsymbol{T}A}$

(K2) $f^* \circ \eta_A = f$

(K3) $g^* \circ f^* = (g^* \circ f)^*$ for $g : B \longrightarrow \boldsymbol{T}C$

[3]Pedicchio and Tholen (2004, p.249-)

Theorem 8.5.2. Every Kleisli triple $\langle \boldsymbol{T}, \eta, -^* \rangle$ over $\mathcal{X}$ has a unique extension to a monad $\langle \boldsymbol{T}, \eta, \mu \rangle$ on $\mathcal{X}$ with:

- $\boldsymbol{T}h = (\eta_B \circ h)^*$ for each morphism $h : A \longrightarrow B$ of $\mathcal{X}$
- $\eta_A = (id_{\boldsymbol{T}A})^*$

Conversely, every monad on $\mathcal{X}$ restricts to a Kleisli triple with $f^* = \eta_B \circ \boldsymbol{T}f$ for each $f : A \longrightarrow \boldsymbol{T}B$ in $\mathcal{X}$.
These processes are inverse to each other.

Exercise 8.5.3. Prove Theorem 8.5.2.

Theorem 8.5.4. The morphism f is an isomorphism in $\mathcal{X}_{\boldsymbol{T}}$ iff f^* is an isomorphism in $\mathcal{X}$.

Exercise 8.5.5. Prove Theorem 8.5.4.

Theorem 8.5.6. Given diagrams

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle g} \\ C & \xrightarrow[k]{} & D \end{array} \qquad \begin{array}{ccc} \boldsymbol{T}A & \xrightarrow{f^*} & \boldsymbol{T}B \\ {\scriptstyle h^*}\downarrow & & \downarrow{\scriptstyle g^*} \\ \boldsymbol{T}C & \xrightarrow[k^*]{} & \boldsymbol{T}D \end{array}$$

it then follows that:

(1) the left square commutes in $\mathcal{X}_{\boldsymbol{T}}$ iff the right square commutes in $\mathcal{X}$.

(2) the left square is a pullback in $\mathcal{X}_{\boldsymbol{T}}$ iff the right square is a pullback in $\mathcal{X}$.

Exercise 8.5.7. Prove Theorem 8.5.6.

Bibliography

Crole, R. L. (1993) *Categories for Types*. Cambridge University Press.

Kleisli, H. (1965) "Every standard construction is induced by a pair of adjoint functors", In the Proceedings of *the American Mathematical Society*, Vol. 16. pp.544–546.

MacLane, S. (1997) *Categories for the Working Mathematician*, Graduate Texts in Mathematics. Springer, second edition edition.

MacLane, S. and I. Moerdijk. (1992) *Sheaves in Geometry and Logic – A First Introduction to Topos Theory –*. Springer.

McLarty, C. (1992) *Elementary Categories, Elementary Toposes*, Oxford Logic Guides 21. Oxford Science Publications.

Moggi, E. (1989) "Computational lambda-calculus and monads", In the Proceedings of *Fourth Annual IEEE Symposium on Logic in Computer Science*. pp.14–23.

Pedicchio, M. C. and W. Tholen. (2004) *Categorical Foundations: Special Topics in Order, Topology, Algebra, and Sheaf Theory*, Vol. 97 of *Encyclopedia of Mathematics and Its Applications*. Cambridge University Press.